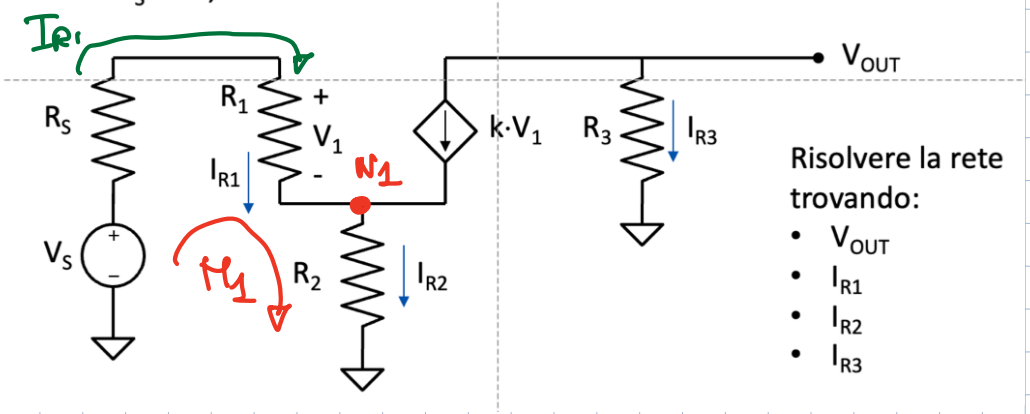
ESERCIZIO GENERATORE PILOTATO

Dati:

$R_S = 1k\Omega$, $R_1 = 1k\Omega$, $R_2 = 2k\Omega$, $R_3 = 3k\Omega$

$V_S = 5V$, $k = 100mS$



Risolvere la rete trovando:

- V_{OUT}
- I_{R1}
- I_{R2}
- I_{R3}

$$M_1: \quad -V_S + R_S I_{R_1} + R_1 I_{R_1} + R_2 I_{R_2} = 0$$

$$V_1 = I_{R_1} R_1$$

$$N_1: \quad I_{R_2} = I_{R_1} + K V_1 = I_{R_1} + K R_1 I_{R_1} = I_{R_1}(1 + K R_1)$$

$$V_S = I_{R_1}(R_S + R_1) + I_{R_1} R_2 (1 + K R_1)$$

$$I_{R_1} = \frac{V_S}{R_S + R_1 + R_2(1 + K R_1)} = \frac{5V}{1k\Omega + 1k\Omega + 2k\Omega(1 + 100mS \cdot 1k\Omega)} = \frac{5V}{204\,k\Omega} = 2{,}45 \cdot 10^{-5} A = 24{,}5\,\mu A$$

101

$$I_{R_2} = I_{R_1}(1 + K R_1) = 101\, I_{R_1} = 2{,}47\ mA$$

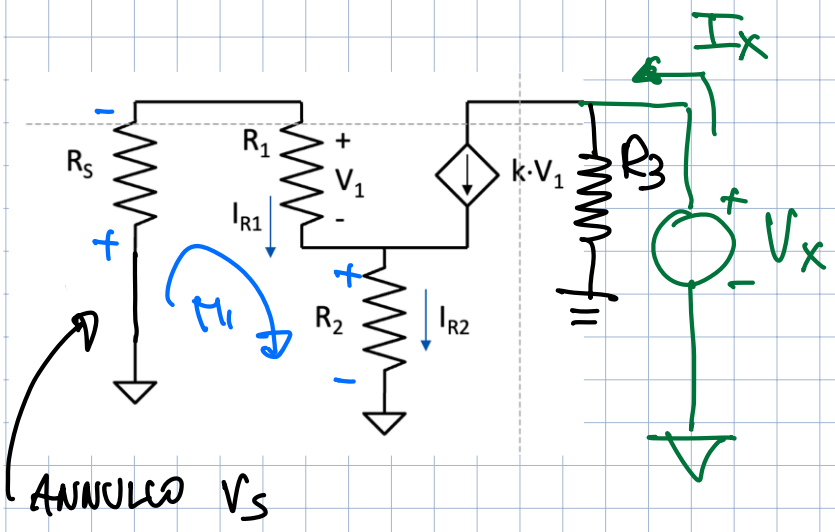
$$I_{R_3} = -K R_1 I_1 = 2{,}45\ mA$$ (NOTARE SEGNO "–")

$$\boxed{V_O = I_{R_3} R_3 = \frac{-K R_1 V_S}{R_S + R_1 + R_2(1 + K R_1)} \cdot R_3}$$

$$V_0 = \frac{-KR_1R_3}{R_S + R_1 + R_2(1+KR_1)} \cdot V_S = -7{,}35\,V$$

$$\Rightarrow \boxed{V_0 = I_{R_3} R_3 = -7{,}35\,V}$$

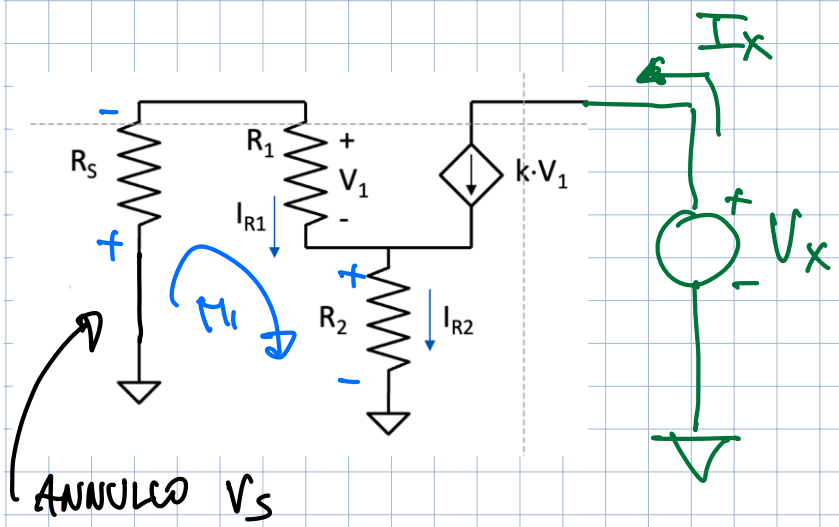
CALCOLO R_{TH}?



$$\Rightarrow R_{TH} = R_3 // R'_{TH}$$

CALCOLO R'_{TH}

$$\boxed{R'_{TH} = \frac{V_x}{I_x}}$$



$$R_S I_{R_1} + R_1 I_{R_1} + R_2 I_{R_2} = 0$$

$$I_{R_2} = I_{R_1} + K R_1 I_{R_1} = (1 + K R_1) I_{R_1}$$

$$I_{R_1} \underbrace{\left[R_S + R_1 + (1+KR_1)R_2\right]}_{\neq 0} = 0$$

$$\Rightarrow I_{R_1} = 0$$

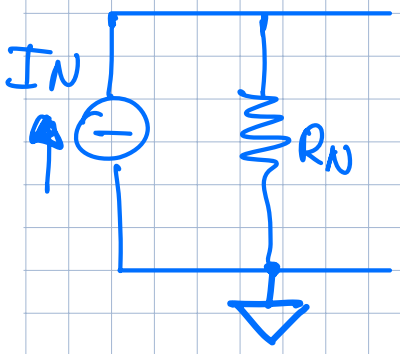
$$\Rightarrow I_x = K V_1 = K R_1 I_{R_1} = 0$$

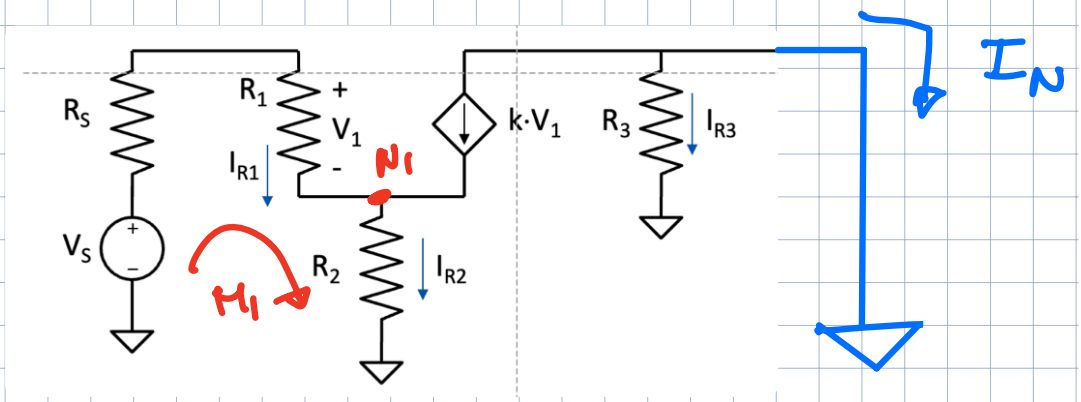
GENERATORE IDEALE

$$\Rightarrow \boxed{R'_{TH} = \infty}$$

$$\Rightarrow R_{TH} = R_3 // \infty = R_3 \;!$$

EQUIVALENTE DI NORTON ?





PER TROVARE I_N USO LO STESSO PROCEDIMENTO DI PRIMA PER CALCOLARE I_{R_3} CON M_1 e N_1:

$$\Rightarrow \quad I_{R_1} = \frac{V_S}{R_S + R_1 + R_2(1+KR_1)} = 24{,}5 \text{ mA}$$

QUIN' $I_N = -KV_1 = -KR_1 I_{R_1} = \dfrac{-KR_1 V_S}{R_S + R_1 + R_2(1+KR_1)}$

$$= -2{,}45 \text{ mA}$$

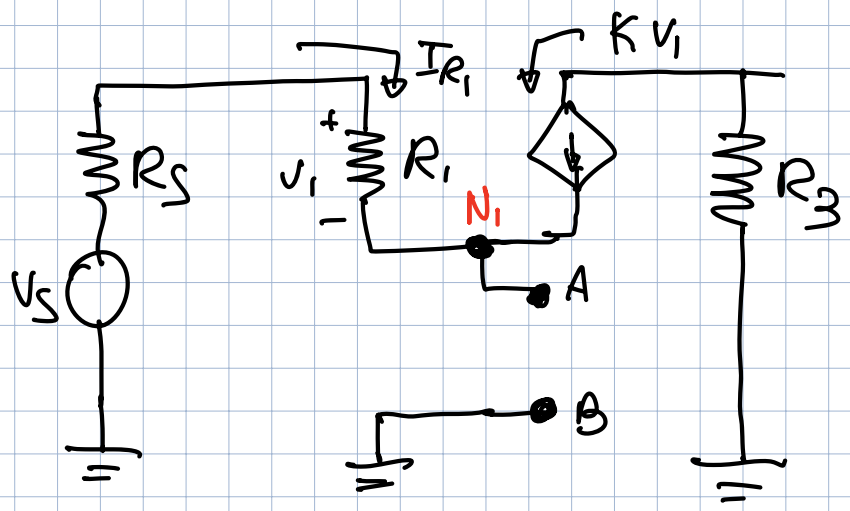
PER IL CALCOLO DI R_N USO PROCEDIMENTO PER CALCOLO DI R_{TH} VISTO CHE $R_N = R_{TH} = R_3$!

TUTTAVIA SO ANCHE CHE:

$$R_N = R_{TH} = \frac{V_{TH}}{I_N} = \frac{\dfrac{\cancel{-}\cancel{KR_1} R_3 \cancel{V_S}}{\cancel{R_S + R_1 + R_2(1+KR_1)}}}{\dfrac{\cancel{-}\cancel{KR_1} \cancel{V_S}}{\cancel{R_S + R_1 + R_2(1+KR_1)}}} = R_3$$

CVD!

EQUIVALENTE DI THEVENIN VISTO DA R_2



NON POSSO USARE LA TAGLIA (Gen. I)

SU N_1:

$$I_{R_1} + K I_{R_1} R_1 = 0$$

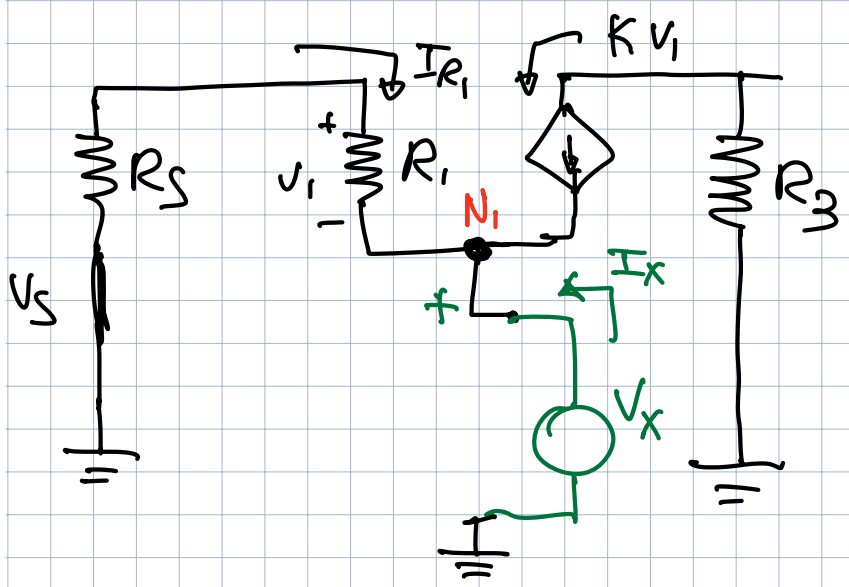
$$I_{R_1} \underbrace{(1 + K R_1)}_{\neq 0} = 0$$

$$\Rightarrow I_{R_1} = 0$$

$$\boxed{\text{SE } I_{R_1} = 0 \Rightarrow V_{AB} = V_S}$$

$$\Rightarrow \boxed{V_{TH} = V_{AB} = V_S}$$

CALCOLO R_{TH}



$$V_X = -I_{R_1}(R_1 + R_S)$$

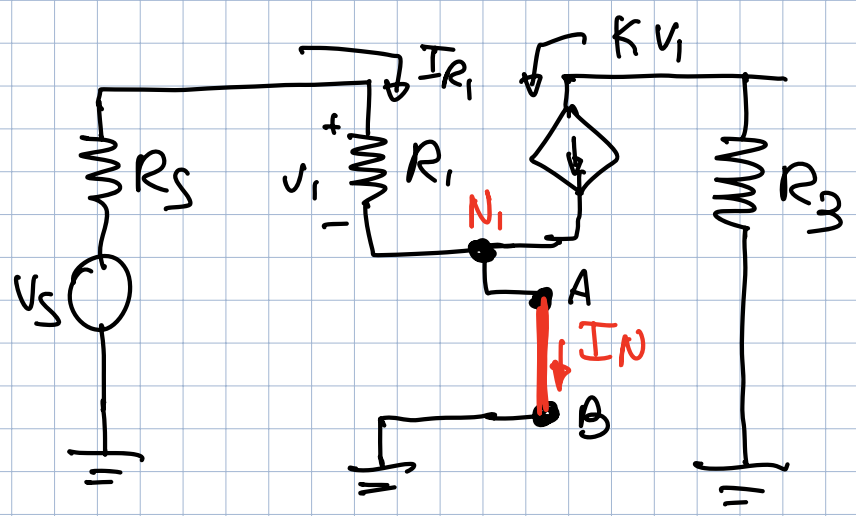
$$I_X = -I_{R_1} - KV_1$$

$$= -I_{R_1} - K R_1 I_{R_1}$$

$$= -I_{R_1}(1 + K R_1)$$

$$\Rightarrow R_{TH} = \frac{V_X}{I_X} = -\frac{I_{R_1}(R_1 + R_S)}{-I_{R_1}(1 + K R_1)} = \boxed{\frac{R_1 + R_S}{1 + K R_1}}$$

EQUIVALENTE DI NORTON VISTO DA R_2 ?



$$I_N = I_{R_1} + KR_1 I_{R_1} = I_{R_1}(1+KR_1)$$

$$I_{R_1} = \frac{V_S}{R_S + R_1}$$

$$\Rightarrow \boxed{I_N = \frac{V_S(1+KR_1)}{R_S + R_1}}$$

$$\boxed{R_N = R_{TH} = \frac{R_1 + R_S}{1+KR_1}}$$

VENI CALCOLO R_{TH} !

INFATTI

$$R_{TH} = R_N = \frac{V_{TH}}{I_N} = \frac{V_S}{\frac{V_S(1+KR_1)}{R_S+R_1}} = \frac{R_S + R_1}{1+KR_1}$$

CVD !